GGG-H-142e
June 6, 1967
SUPERSEDING
Int. Fed. Spec. GGG-H-00142d(NAVY-Ships)
June 15, 1966 and
Fed. Spec. GGG-H-142c
December 18, 1961

FEDERAL SPECIFICATION

HELMET, CONSTRUCTION WORKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of safety helmets for protection of the wearer's head against impact, penetration, and electrical shock.

1.2 Classification.

1.2.1 Types. The helmets shall be furnished in the following types, as specified (see 6.2).

Type I Hat, (full brim).
Type II Cap, (brimless with peak).

1.2.2 Colors. Helmets shall be furnished in the colors listed in 3.5, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 406 - Plastics: Methods of Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which include cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The helmets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Components. Each helmet shall consist of a shell, suspension (headband and crown straps forming an internal cradle for supporting the helmet on the wearer's head,), a sweatband, and when specified, shall include an adjustable chinstrap and a winter liner (see 6.2).

3.3 Materials. Materials used in the fabrication of the helmet shells furnished under this specification shall be at the option of the manufacturer. Material coming into contact with the wearer's head shall not be a type known to cause skin irritation or disease.

3.4 Construction.

3.4.1 Shell. The shell shall be of one-piece seamless construction, dome shaped to provide the inside clearances specified herein.

3.4.1.1 Metal parts. Metal parts shall not impair the electrical protection which the shell provides for the wearer. They shall be so placed that no part of the worker's head is likely to be scratched by a blow on the brim of the helmet which would cause the helmet to slide down on the ears or forehead. Metal parts shall be rust resistant.

3.4.1.2 Brim.

3.4.1.2.1 Type I, full brim. The brim for type I helmets shall be an integral extension of the shell dome, and shall be continuous around the helmet. The width of the brim shall be not less than 1-1/2 inches and not more than 3 inches measured from the inside edge of the shell except that the width at the front may taper to not less than 1-1/4 inches. With the helmet held in a horizontal position, the brim shall slope downward in a straight line at an angle of not less than 15 degrees nor more than 37 degrees from the horizontal.

3.4.1.3 Type II, brimless with peak. The peak for type II helmets shall extend forward from the crown not less than 1-1/2 inches, and shall be not less than 5-1/2 inches wide at the apex. Type II helmets shall not be made by cutting the brim from type I helmets.

3.4.2 Headbands. Headbands shall be adjustable and shall permit adjustment to all sizes within the range marked on helmet and band. The headband (which may be a replaceable component) shall be securely attached either to the helmet shell or to the crown straps for assuring its proper position in the helmet. The manner of the attachment shall not intefere with the adjustments of the crown straps.

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3.4.2.1 Sweatband. Sweatbands of the removable-replaceable type shall be provided or if sweatband is integral with headband, the headband shall be a replaceable component of the suspension assembly. The sweatband shall cover not less than the forehead portion of the headband and shall be at least as wide as the headband.

3.4.2.2 Adjustment range. Headbands shall be adjustable in 1/8-size increments from at least size 6-3/4 to 7-1/2, inclusive. The sizes shall be such that the circumference of the inside of the headband for size 7 is 22 inches, with a 3/8-inch increment for each 1/8 size above and below size 7. Tolerance of circumferential measurements shall be ± 1/8 inch.

3.4.2.3 Headband clearance. As determined by the method described in 4.5.8 the clearance between the inside of the shell and the outside of the headband shall be not less than 1/4 inch.

3.4.3 Crown straps. Crown straps shall be not less than 1/2 inch in width. These straps, when adjusted and properly interlaced, shall form a cradle or hammock for supporting the helmet on the wearer's head and holding the helmet away from the head, thus acting as a shock absorber when the helmet is struck. The number of straps and their location in the helmet shall be suitable for the above purpose. Crown straps shall be so designed that the clearance between the standard head form and the shell of the hat or cap cannot be adjusted to less than the 1-1 inches minimum clearance requirement permitted in 4.5.3.4 (b).

3.4.4 Accessories. If chinstraps and winter liners are specified in the contract or order (see 6.2), they shall conform to the following requirements:

3.4.4.1 Chinstrap. The chinstrap shall be adjustable and made of closely woven webbing, leather, elastic cotton webbing combination, or plastic, of not less than 1/2 inch in width.

3.4.4.2 Winter liner. The winter liner shall be designed to cover the skull, neck, and ears. The skullcap, part of the winter liner, shall be attached to the helmet cradle or to the helmet in such a manner that it can be readily removed and reattached. The earlug and neck part may be made either in one piece with the skullcap or attachable to it by tabs or a zipper, as specified. The neck and earlug shall be drawn close to the face by a lace or a chinstrap. The winter liner shall consist of fabric, plastic, or any suitable material; colored material shall be fast dyed. The outer surface may be water repellant.

3.5 Colors. When specified by the procuring agency (see 6.2), helmets shall be furnished in color. Colors shall conform to Fed. Std. No. 595 for the color number specified. Department of Defense purchases shall be restricted to the following standard colors:

Color	Color Number (Fed. Std. No. 595)*
Maroon	10075
Terra Cotta	10233
Red	11105
Orange	12197
Yellow	13538
Green	14260
Blue	15177
Gray	16492
Black	17038
White	17875

*Individual (3 by 5 inch) color chips are available from General Services Administration, Business Service Center, Washington, D. C.

3.5.1 Color application. The color shall be provided by the use of high-quality pigment, pigmented lacquers, alkyd enamel or polyester finish. Injection molded shells shall be pigmented with applicable color during molding. The use of surface colorants shall not be permitted.

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3.6 Physical requirements.

3.6.1 Insulation resistance. The helmet shall not show a leakage current in excess of 3 milliamperes when tested as specified in 4.5.2.

3.6.2 Impact resistance. When tested in accordance with 4.5.3, helmets shall not transmit an average force of more than 850 pounds. Individual specimens shall not transmit a force greater than 1,000 pounds. No part of helmet or suspension shall break under impact test.

3.6.3 Penetration strength. The helmet shall be neither dented nor pierced for more than 3/8 inch, nor shall the helmet be forced down over the standard head form, nor shall the straps pull out or break when tested as specified in 4.5.4.

3.6.4 Edge strength, type II. When tested in accordance with 4.5.5, type II helmets shall show a deflection under a 40 pound load of not more than 3/4 inch and an ultimate strength of not less than 50 pounds.

3.6.5 Flammability. The thinnest section of the shell shall not ignite in less than 5 seconds and shall not burn at a rate greater than 3 inches per minute when tested as specified in 4.5.6.

3.6.6 Water absorption. The shell material of the helmet shall absorb not more than 5 percent of water when tested as specified in 4.5.7.

3.6.7 Weight. The weight of each complete helmet without winter liner and chinstrap or other accessories shall not exceed 15 ounces (see 4.5.9).

3.7 Interchangeability. The suspension shall be easily removable and replaceable and shall be interchangeable among shells of the same size and model furnished by the same manufacturer (see 6.2).

3.8 Instructions. Instructions which explain the proper method of wearing and adjusting the helmet shall be furnished with each helmet. They shall be stowed in the helmet so as to remain where placed until intentionally removed.

3.9 Marking.

3.9.1 Shell. The inside of each helmet shell shall be marked in a permanently legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.9.2 Headband. The range of sizes to which the headband is adjustable shall be marked on labels cemented to, or embossed upon, the helmet shell or the headband.

3.10 Workmanship. The workmanship shall be first class in every respect. The safety helmets shall have no sharp edges or burrs, and shall be free from any other defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the visual and dimensional examination of 4.4.1 and the tests specified in 4.5, except that the test specified in 4.5.5 applies to type II helmets only. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.4).

4.2.1 Samples for qualification tests. (Thirty-six) helmets of each type and material for which approval is desired shall be furnished. Qualification will cover only helmets of the material from which the sample was made.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All helmets of one type presented for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sampling for visual and dimensional examination. A random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level III. The acceptable quality level shall be 2.5 defects per hundred units for major defects and 4.0 defects per hundred units for minor defects.

4.3.3 Sampling for insulation-resistance test. A random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-4. AQL shall be 1.5 defects per hundred units (see 4.5.2).

4.3.4 Sampling for impact, penetration, and edge strength (type II). From each lot, a random sample of helmets shall be taken separately for each test in accordance with MIL-STD-105 at inspection level S-2. AQL shall be 1.0 defects per hundred units (see 4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample helmet selected in accordance with 4.3.2 through 4.3.4. Quality conformance inspection shall consist of the following:

(a) Examination (4.4.1).
(b) Tests (4.4.2).
(c) Inspection of preparation for delivery requirements (4.6).

4.4.1 Visual and dimensional examination. Each of the sample helmets taken in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as indicated in table I. Any helmet in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective helmets in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

TABLE I

Examine	Defect	Major	Minor
Finish	Color or type of finish not in accordance with approved sample or as required - - - -	X	
Design	Any characteristic not in accordance with the specified requirements or approved sample seriously affecting serviceability - - -	X	
	Any characteristic not in accordance with the specified requirements of approved sample affecting serviceability but not seriously -		X

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TABLE I (cont'd)

Examine	Defect	Major	Minor
Material (general)	Any component not fabricated of approved material seriously affecting serviceability.	X	
	Any component not fabricated of the approved material not seriously affecting serviceability, e.g., rivet is fabricated from steel when brass is specified - - - - - - -		X
Headbands or chinstraps	Side in contact with skin not smooth - - -		X
	Any sharp burr, sliver, or edge - - - - - - -		X
	Any cuts, holes, open scratches, wrinkles, or scars affecting serviceability but not seriously - - - - - - - - - - - - - - - - - -		X
	Spots or stains that are clearly noticeable.		X
	Any assembly that is not adjustable where required, e.g., chinstrap not adjustable - - -	X	
	Any component or assembly malformed seriously affecting serviceability. - - - - - - - - -	X	
	Any component or assembly malformed affecting serviceability but not seriously - -		X
	Any fastener, i.e., eyelets, rivets missing or not securely set to a degree where it may become detached from assembly - - - - -	X	
	More than one fastener not securely set but is not expected to become detached from assembly -		X
	Any fastener too tightly set or compressed resulting in damage to assembly, e.g., too much pressure applied to rivet resulting in damage to shell - - - - - - - - - - - - - -	X	
Assembly highlights	Attachment of the cradle assembly to helmet does not provide adequate clearance between the shell and the headband as compared with approved sample or requirement -	X	
Crown straps and laces	Any rip, hole, cut, tear, or other defect seriously affecting serviceability - - - -	X	
	Defective to a degree where serviceability will be affected but not seriously, e.g., misweave, slubs, etc., if applicable. - - - - -		X

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TABLE I (cont'd.)

Examine	Defect	Major	Minor
Shell	Any hole, soft spot, air pocket, void, evidence of delamination, rough or abraded area, cracked, chipped, blistered, pitted or otherwise defective, seriously affecting serviceability -	X	
	Scratches, air pocket, void, rough or abraded area, chipped, pitted, blistered or otherwise defective, affecting serviceability but not seriously, or seriously affecting appearance -		X
	Metal parts and metal clips for attaching crown straps and headbands to shell, positioned in manner likely to cause injury if blow on brim forces helmet over ears or forehead - - - - - - - - - - - - - - - - - -	X	
Construction and workmanship general (applicable to all components, unless otherwise indicated herein)	Any component omitted, damaged, misplaced, operation omitted or not properly performed seriously affecting serviceability - - - -	X	
	Any component misplaced, damaged, operation not properly performed affecting serviceability but not seriously - - - - - - -		X
	Any component not firmly affixed or attached to an extent that parts or assembly may become separated in use - - - - - - - - - - - -	X	
	Any component not firmly affixed or attached but part or assembly is not expected to become detached in use - - - - - - - - -		X

4.4.2 Quality conformance tests. Sample helmets taken in accordance with 4.3.3 and 4.3.4 shall be subjected to the tests specified in 4.5.2 through ~~4.5.5.3~~ and 4.5.9. Sample helmets failing one or more of these tests shall not be offered for delivery. If the number of defective helmets in any sample exceeds the acceptance number for the samples tested, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 Preparation of samples. Where it is evident that the sample helmets under test have a protective coating over the basic material, the entire exterior surface of the shell shall be abraded until the basic material is exposed using number 60-grit garnet paper. All samples shall be conditioned in an oven for 96 hours at 50° ± 2° C., then cooled in a desiccator and exposed for 96 hours in an atmosphere having 50 ± 2 percent relative humidity and a temperature of 25° ± 2° C. The samples shall then be subjected to the applicable tests specified in 4.5.2 through 4.5.5.

4.5.2 Insulation-resistance test. The insulation-resistance test shall be performed as follows to determine conformance with 3.6.1:

4.5.2.1 Apparatus. The test apparatus shall consist of the following:

(a) A vessel containing tapwater of sufficient size to submerge an inverted helmet to within 1/2 inch of the junction of brim with the dome.

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(b) Wire frame for suspending inverted helmet in tapwater.

(c) Source of 60-cycle alternating current with 2,200 volts (root mean square).

(d) Wiring and terminals for application of voltage across dome of helmet.

(e) Voltmeter of sufficient capacity.

(f) Millimeter of sufficient capacity and accuracy to measure the specified current.

4.5.2.2 Procedure. An inverted helmet with the suspension removed shall be placed on the wire frame and submerged in the tapwater to within 1/2 inch of the junction of the brim with the dome. The helmet shall also contain the tapwater to the same level. If the helmet is provided with holes in the crown near the brim, the helmet shall be submerged to within 1/2 inch of the holes. One terminal from the current source shall be connected to the wire frame which supports the helmet and the other shall be immersed in the tapwater which is contained by the helmet. The 2,200 volts shall then be applied for 1 minute. If passage of current is evidenced by deflection of the voltmeter the helmet shall be considered as having failed to conform to 3.6.1. If no deflection of voltmeter occurs, a millimeter is placed in the circuit and a reading taken.

4.5.2.3 Report. For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.5.3 Impact test. The impact test shall be performed as follows to determine conformance with 3.6.2:

4.5.3.1 Apparatus. The test apparatus shall consist of the following:

(a) A standard head form (see 6.5). The standard head forms are of four sizes: Small, medium, large, and extra large, representing head sizes 6-3/4, 7, 7-1/4 and 7-1/2, respectively. Test-block forms which simulate the standard head forms may be employed. The blocks shall duplicate the crown dimensions and contours from the top down to the ear and eye levels, including the top portion of the ears. To protect the head form, it may be provided with a steel insert (see fig. 1). For the clearance, impact and penetration resistance test the headband should be adjusted to fit similar to the fit obtained when worn on a man's head and the hat or cap mounted on the 7-~~3/4~~ size standard head form. Where the headband cannot be adjusted to fit the 7-1/4 size standard head form, the hat or cap shall be mounted on the appropriate size of standard head form.

(b) A Brinell penetrator assembly as shown on figure 1. The impression bar shall be of metal having a predetermined Brinell hardness of 18 to 30 as measured with a 500-kilogram load using a 10-millimeter ball. The thickness of the bar shall be such that it is at least 10 times the depth of any indentation and the width such that the distance from the center of any indentation to the edge of the bar is at least 2-1/2 times the diameter of the indentation. (An impression bar made of S-O aluminum 1/2 inch thick and 1-1/8 inch wide fulfills these conditions within the specified load range.) The Brinell penetrator used in the impact test shall be a hardened-steel ball 1/2 inch in diameter.

(c) A steel ball approximately 3-3/4 inches in diameter and weighing 7.8 to 8.0 pounds.

(d) A Brinell microscope or other suitable micrometer microscope accurate to 0.05 millimeter (mm).

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4.5.3.2 Test conditions.

4.5.3.2.1 For qualification tests. Impact tests shall be conducted with the specimens maintained at temperatures of 20° ± 2° F. and 120° F. Specimens shall be subjected to the test temperature for at least 4 hours prior to the impact test.

NOTE: Elapsed time from removal of helmet from controlled temperature to time of impact shall not exceed 15 seconds.

4.5.3.2.2 For quality conformance tests. Impact tests shall be conducted with the specimens maintained at the prevailing atmospheric temperature.

4.5.3.3 Size of specimens (for qualification tests). Specimens shall be tested on the 7 1/4 size standard head form except as noted in 4.5.3.1 (a).

4.5.3.4 Procedure.

(a) Mounting specimens. Specimens shall be properly centered on the appropriate head form, in a position similar to that when worn on a man's head, with the back of the hat or cap towards the fulcrum of the impact test base assembly.

(b) Crown clearance. The clearance shall be determined by placing the hat or cap on the appropriate size of standard head form. The crown adjustment laces, if any, shall be removed. A 25-pound weight with a flat surface of not less than 2 inches in diameter shall be placed on the shell, and the distance between the underside of the shell and the crown of the head form shall be considered to be the clearance. To determine the clearance, the shell, minus suspension, shall be placed on the head form and a dimensional reading taken as described (see fig. 2). The suspensions shall than be installed and another dimensional reading taken with the 25-pound-weight applied to the crown of the shell. The difference in dimensional readings shall be the clearance. The clearance between the shell of the specimen and the standard head form shall be not less than 1-1/4 inches minimum (measured with the tie cord out if the assembly uses a tie cord) under a 25 pound load.

(c) Application of impact. The Brinell penetrator assembly shall be placed on a level surface that will not deflect during the test. The drop ball, the head form and the penetrator ball shall be center aligned by means of a plumb bob. For testing, the center of the crown of the hat or cap shall be as nearly centered as possible by adjustment of the headband. The hat or cap shall be mounted with the back towards the fulcrum. The clearance shall be adjusted to the minimum clearance determined in accordance with 4.5.3.4 (b). The ball shall be dropped vertically immediately after aligning the standard head form as required, from a height of 60 inches, measured from the bottom of the ball to the top of the highest point on the shell. An electromagnetic release is suitable. The ball shall not be allowed to strike the hat or cap more than once. Impressions on the bar shall be spaced at least 2-1/2 diameters apart, edge to edge. The diameter of impression on the test bar shall be measured and recorded to the nearest 0.1 millimeter using the Brinell microscope. Elliptical impressions shall be disregarded if the difference between the minimum and maximum axes exceeds 0.3 millimeters. All impressions from double blows shall be disregarded. For each test, the average Brinell hardness number of the 7-inch impression bar used shall be recorded.

4.5.3.5 Report. The average force and the greatest individual force for all helmets tested under one group shall be computed and reported. The forces shall be computed from the impression diameters using the following Brinell formula:

$$F = 2.2 \times H \times \frac{\pi D}{2} \times \left(D - \sqrt{D^2 - d^2}\right)$$

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Where:

F is the transmitted force in pounds,
H is the average Brinell hardness number of the impression bar,
D is the diameter of the impression ball in millimeters, and
d is the diameter of the impression in millimeters (see 6.6).

4.5.4 Penetration test. The penetration test shall be performed as follows to determine conformance with 3.6.3:

4.5.4.1 Apparatus. The test apparatus shall consist of a (7 1/4) size standard head form or test block as specified in 4.5.3.1 (a) and a 16-ounce plumb bob with a steel point having an included angle of 30° 1' and a maximum point radius of 0.010 inch.

4.5.4.2 Procedure. The specimen shall be mounted on the apparatus head form as specified in 4.5.3.4 (a) and the plumb bob dropped onto the geometric center of the crown within a 3 inch diameter circle from the height of 10 feet, measured from the bottom point of plumb bob to the top of the helmet. The depth the plumb bob has pierced the helmet shall be measured to the nearest 1/32 inch and shall include the thickness of the shell material. Plumb bob shall not fall on ridges or the injection point.

4.5.4.3 Report. For each specimen, the depth of penetration to the nearest 1/32 inch shall be reported.

4.5.5 Edge-strength test, type II. The edge-strength test shall be performed as follows on type II helmets to determine conformance with 3.6.4:

4.5.5.1 Apparatus. The edge-strength test apparatus shall consist of a fixture similar to figure 3 and capable of applying a load of at least 50 pounds. The apparatus shall consist of a fabricated metal frame incorporating a gaging and weight rod bearing support, a gaging and weight rod, a dial test indicator, and the necessary weights to provide loads in 10-pound increments up to 50 pounds. The gaging weight rod shall include a metal disk supporting the load weights and resting on a thrust bearing. The thrust bearing shall have a 4-inch diameter disk attached to the bottom race and to the gaging rod. Disk shall be grooved to accept helmets with roll or flared edges around the rim.

The dial test indicator shall have at least a 3-inch range and shall be positioned so that it can be aligned with a disk mounted on top of the gaging rod. A right angle wood fixture suitable for mounting helmets, which shall be supported in position by a rubber "O" ring in the manner shown on figure 3 shall be included. "O" ring diameter and length shall be such as to hold the helmet in position for application of the test load.

4.5.5.2 Procedure. With the helmet mounted in right angle wood fixture, the gaging rod with metal disks and thrust bearing is placed and centered on the helmet rim. All weight is removed from the helmet and with the bottom disk in contact with the helmet the dial indicator is set at zero with at least a 3-inch spindle travel preset. The gaging rod is released and sufficient weight added to reflect dead weight of 20, 30 and 40 pounds. The deflection as indicated on the dial indicator shall be recorded for each of the above weights. A 50 pound load shall be applied to test for ultimate strength requirement of 3.6.4.

4.5.5.3 Report. Each deflection under loads of 20, 30 and 40 pounds for each sample tested shall be recorded. Failure of helmet at less than an ultimate weight of 50 pounds shall be recorded.

4.5.6 Flammability test. The flammability test shall be performed as follows to determine conformance with 3.6.5:

4.5.6.1 Apparatus. The test apparatus shall consist of the following:

(a) Rigid support and clamps for mounting test strips in specified position.
(b) Alcohol lamp or gas burner.

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(c) Stopwatch
(d) An enclosed flame resistant, vented chamber with a door containing heat-resistant glass window.

4.5.6.2 Test conditions. Test specimens shall be subjected to the flammability test after subjection to the tests specified in 4.5.3, 4.5.4 and 4.5.5 when applicable.

4.5.6.3 Procedure. Three strips approximately 1/2 inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner.

These strips shall be cut from a portion of the hat so as to give as flat a section as practicable. Lines shall be drawn across each strip at 1/2-inch intervals, measured along the surface and starting from one end. The other end shall be clamped to the rigid support so that the logitudinal axis is horizontal and the transverse axis is inclined 45°. The support shall be placed in the test chamber. The alcohol lamp or gas burner, with a neutral, blue flame 1/2 to 3/4 inch in height, shall be placed under the free end of the strips, so adjusted that the flame tip is just in contact with the material. A stopwatch shall be started when the flame is placed in contact with the strip under test. The time required for the material to ignite shall be recorded. At the end of 30 seconds the flame shall be removed and the strip allowed to burn, noting the rate of burning.

4.5.6.4 Report. The ignition time reported and the burning rate reported shall be the average for the three specimens tested.

4.5.7 Water absorption test. Method 7031, immersion procedure "A" of Fed. Std. No. 406 shall be employed for the water-absorption test to determine conformance with 3.6.6. Three pieces, 1 by 3 inches, shall be cut from the shells of the hats prepared in accordance with 4.5.1 and after subjection to one of the tests specified in 4.5.3, 4.5.4 or 4.5.5. Water absorption reported shall be the average of that determined for the three specimens.

4.5.8 Headband clearance measurement. To determine conformance with 3.4.2.3, clearance between the headband and helmet shall be determined by measuring the circumference of the outside of the headband (at the maximum required size for adjustable headbands in accordance with 3.4.2.2) and of the inside of the shell at the juncture of the dome and the visor and computing the diameters and clearance by the following formulas:

$$\text{Diameter} = \frac{\text{Circumference}}{\pi}$$

$$\text{Clearance} = \frac{D-d}{2}$$

D - inside diameter of shell.
d - outside diameter of headband (maximum size).

4.5.9 Weight. Helmets shall be weighed complete without winter liner, chinstrap or other accessories on any suitable weighing balance to the nearest tenth (0.1) ounce to determine conformance with 3.6.7.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

TABLE II

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.

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TABLE II (cont'd.)

Examine	Defect
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, improper taping, etc.
	Bulging or distortion of containers.
Content	Number per bundle is more or less than required.
Weight	Net weight of contents more than maximum specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each helmet shall be wrapped in kraft wrapping paper. The paper shall be molded over the crown of the helmet, extend around the brim, and be tucked on the inside of the crown. Five wrapped helmets shall be stacked together by nesting the crowns one within the other.

5.1.2 Level C. Unless otherwise specified (see 6.2), helmets shall be packaged according to the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Each shipping container (level A or B) shall be provided with a full height slotted partition made of double-faced corrugated fiberboard which shall divide the box into four equal sections. The container and inner packing shall be designed so that a stack of five packaged helmets shall fit snugly in the partitioned section of the box.

5.2.2 Level A. Helmets packaged as specified in 5.1.1 and 5.1.2, shall be packed in snug-fitting, wood-cleated fiberboard, wood-cleated plywood, nailed wood or wire-bound boxes conforming to PPP-B-591 (class II), PPP-B-601 (overseas type), PPP-B-621 (class 2) and PPP-B-585 (class 3), respectively. Containers shall be lined with a waterproof caseliner conforming to MIL-L-10547 sealed in accordance with the appendix thereto. Shipping containers shall be closed and strapped. The gross weight of boxes shall not exceed 150 pounds.

5.2.3 Level B. Twenty helmets packaged as specified in 5.1.1 shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF or SF, class domestic, variety SW or DW. The box shall be closed in accordance with the appendix to the box specification.

5.2.4 Level C. Unless otherwise specified (see 6.2), helmets packaged as specified in 5.1.2 shall be packed in a manner which will insure acceptance and safe delivery at destination. Containers shall comply with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to marking required in the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123 (see 6.2).

5.3.2 Military agencies. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

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6. NOTES

6.1 Intended use. Helmets covered by this specification are intended to protect the head and neck from being struck by falling objects or from being struck against objects that may cause injury. This protection is especially necessary in shipbuilding and repair, steel erection, bridge, road and air field construction, mining, stevedoring, tunneling, aircraft overhaul and maintenance, building construction, timber work and manufacturing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) If adjustable chinstrap and winter liner are required (see 3.2 and 3.4.4).
(d) Color and color number, when required (see 1.2.2 and 3.5).
(e) Quantity of each required.
(f) Level of packaging, if different (see 5.1.2).
(g) Level of packing, if different (see 5.2.4)
(h) Levels of packaging and packing required (see 5.1 and 5.2).
(i) Marking required (see 5.3).

6.2.1 When suspensions or components of the suspensions (for replacement purposes) are ordered, the manufacturer's type or model number of the helmet for which the replacement suspension or component is ordered should be stated, or if required, a representative helmet should be furnished in order that the manufacturer may have the necessary identifying information.

6.2.2 For civil agency procurement. When Level B packaging is required for civil agency procurement the level A requirements in 5.1.1 shall apply.

6.3 With respect to products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL GGG-H-142. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Commander, Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.4).

6.4 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Standard head forms. The standard head forms mentioned throughout this specification are the model heads known as "A.M.L. Head Size Standards," a set of which is located at each of the following places:

National Bureau of Standards
Washington, D. C. 20234

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio

Nondestructive and Tool Testing Division
Laboratory Branch
Quality Assurance Section
Philadelphia Naval Shipyard
Philadelphia, Pennsylvania 19112

6.6 For convenience, table III shows the transmitted forces for the measured impression diameters on impression bars ranging in Brinell hardness from 18 to 30 as computed from the Brinell formula (see 4.5.3.5).

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MILITARY CUSTODIANS:

Army - GL
Navy - SH
Air Force - 82

Review activities:

Army - GL, MD
Navy - SA, SH, YD
Air Force - 82, 85

User activity:

Army - ME

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS
INT-BR
TR-CG
LABOR

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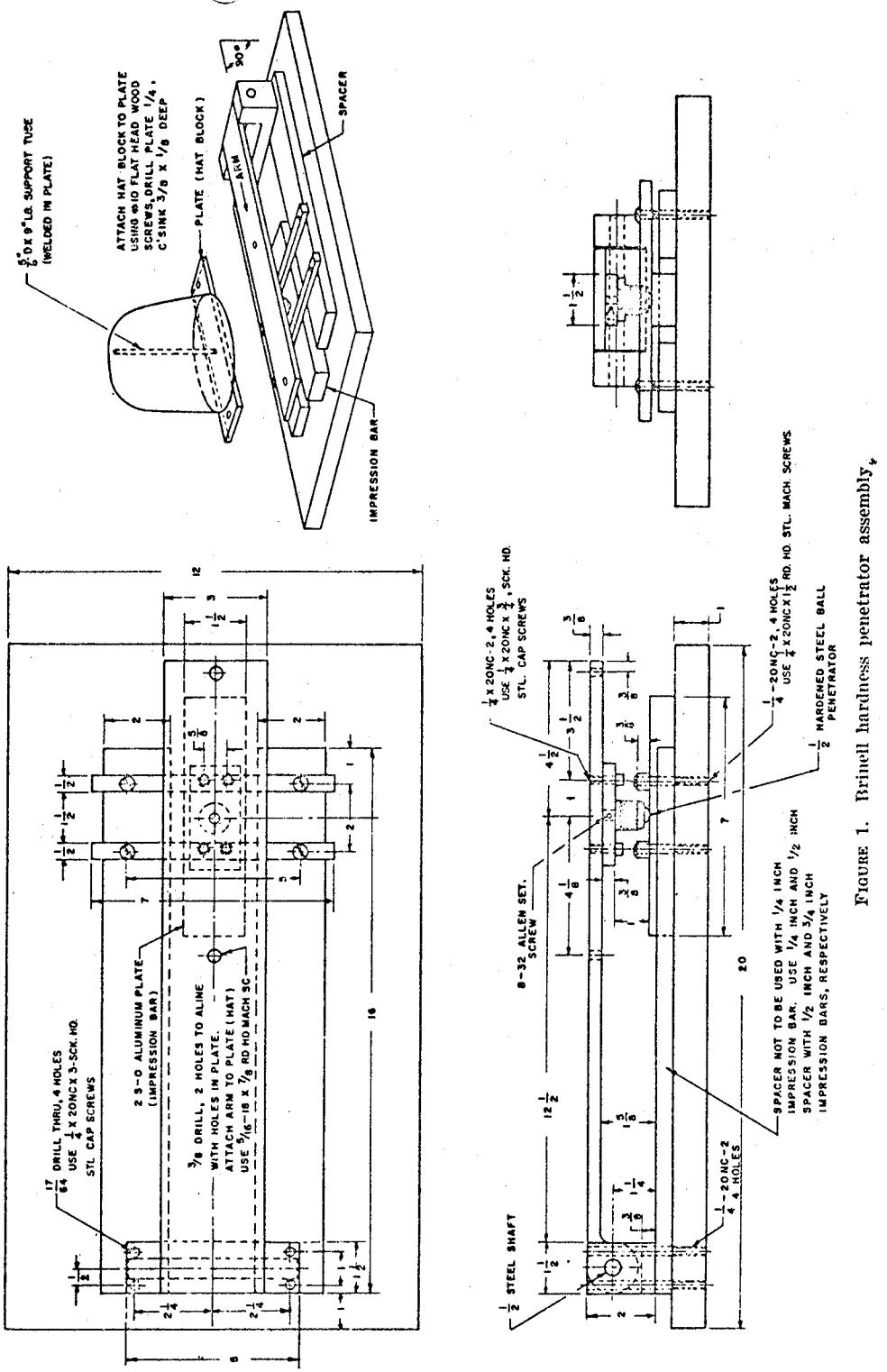


FIGURE 1. Brinell hardness penetrator assembly.

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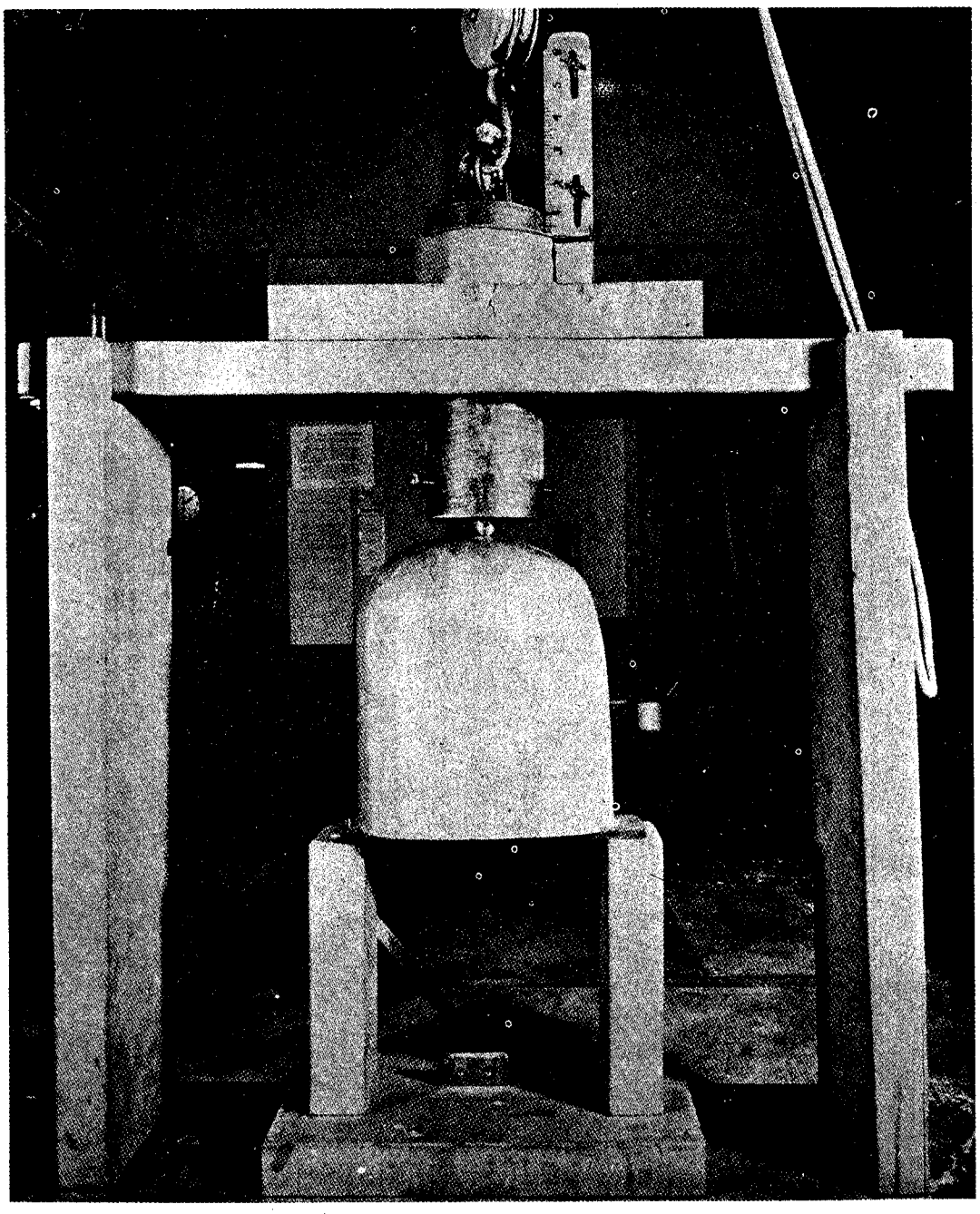


FIGURE 2. Suggested apparatus for measurement of crown clearance (see 4.5.3.4 (b)).

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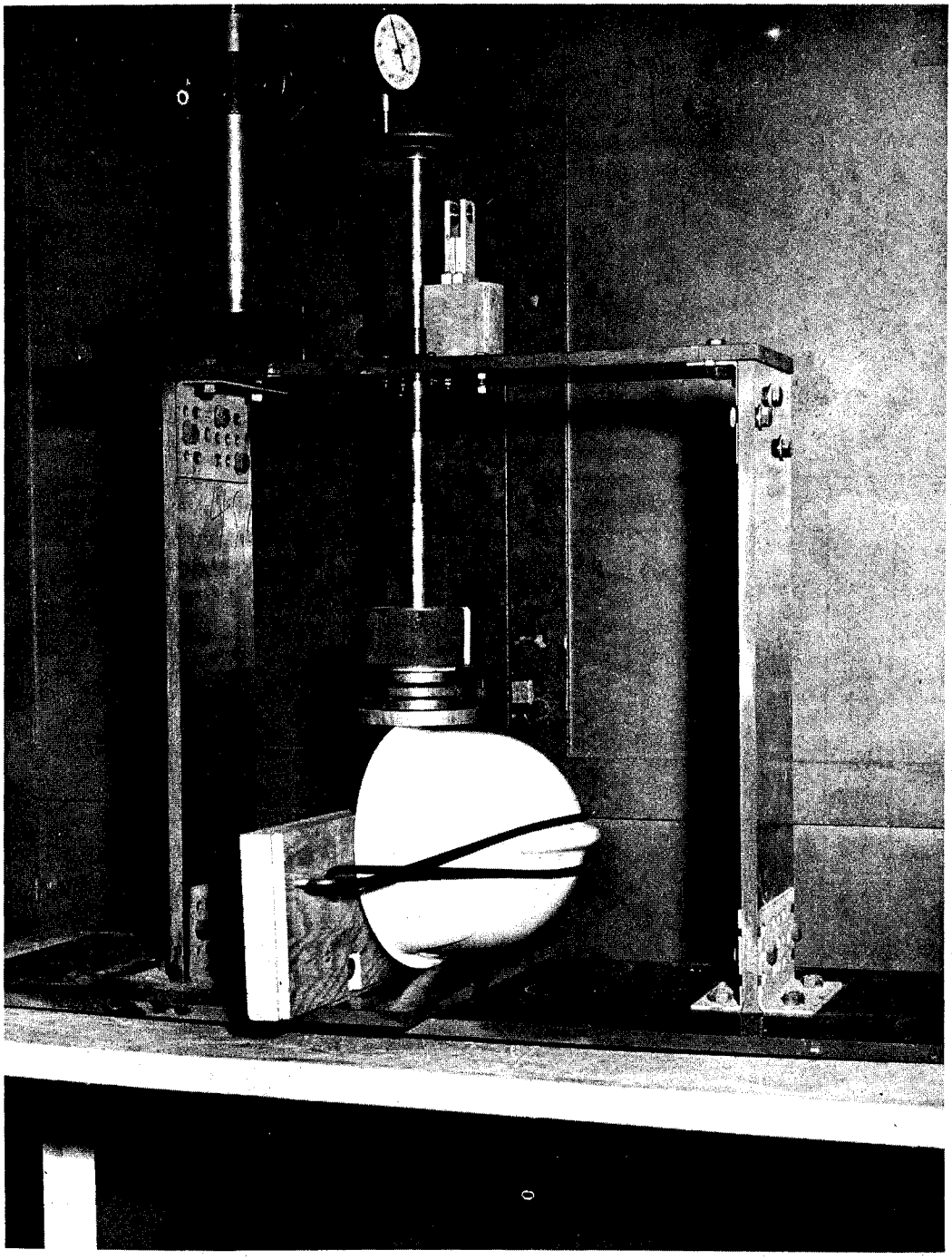


FIGURE 3. Arrangement of apparatus and sample in edge-strength test.

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TABLE III. Transmitted forces in pounds

Diameter of impression, millimeters	Brinell hardness number												
	18	19	20	21	22	23	24	25	26	27	28	29	30
3.9	485	515	540	565	595	620	650	675	700	730	755	785	810
4.0	510	540	570	600	625	655	680	710	740	765	795	825	850A
4.1	540	570	600	630	660	690	720	750	775	810	835	865	895
4.2	565	600	630	660	690	720	755	785	815	850	880	910	940
4.3	595	625	660	695	725	760	790	825	860	890	925	955	990B
4.4	625	660	690	725	760	795	830	865	900	935	970	1000	1040
4.5	650	690	725	760	795	835	870	905	940	975	1015	1050	1085
4.6	680	720	760	795	835	870	910	950	985	1025	1060	1100	1135
4.7	715	755	790	830	870	910	950	990	1030	1070	1110	1150	1190
4.8	745	785	825	870	910	950	990	1035	1075	1115	1155	1200	1240
4.9	780	820	865	905	950	995	1035	1080	1125	1165	1210	1255	1295
5.0	810	850	900	945	990	1040	1080	1130	1175	1220	1265	1310	1355
5.1	A845	895	940	985	1035	1080	1130	1175	1220	1270	1315	1365	1410
5.2	880	930	980	1030	1080	1125	1175	1225	1275	1325	1370	1420	1470
5.3	920	970	1020	1070	1120	1175	1225	1275	1325	1375	1430	1480	1530
5.4	955	1005	1060	1115	1165	1220	1270	1325	1380	1430	1485	1535	1590
5.5	B990	1045	1100	1155	1210	1265	1320	1375	1430	1485	1540	1595	1650
5.6	1030	1085	1140	1200	1255	1310	1370	1425	1480	1540	1600	1655	1710

Notes: Values below line A are above or equal to the specified average force.
Values below line B are above the specified individual force.

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